Advanced Antenna Techniques and Measurements

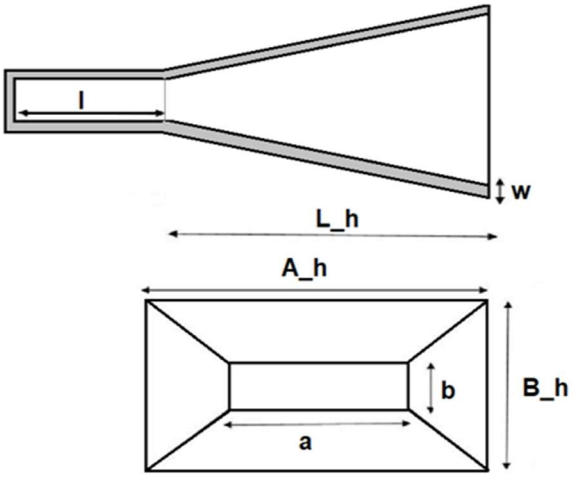
Final assignment

With your assigned group partner(s), write a report that addresses the following questions.

Part 1: Simulation and measurement of two standard gain horns

The first part of the assignment is a study of two horn standard gain horn (SGH) antennas at the Ka-band and C-band, with the parameters specified in the below table.

		Ka-band Horn	C-band Horn
Frequency range		26.5 – 40 GHz	3.95 – 5.85 GHz
Symbol	Description	Value [mm]	Value [mm]
a	Waveguide width	7.112	47.55
b	Waveguide height	3.556	22.2
l	Waveguide length	31.5	68
w	Wall thickness	1	8.4
A_h	Aperture width	68.5	216
B_h	Aperture height	56.5	160
L_h	Length of the horn	150	240



First, the following data of the antennas must be produced:

1. The *approximate* theoretical far-field of the SGHs is obtained from Balanis Eqs. (13-46b), (13.46c) .
2. The simulation far-fields are obtained from the HFSS models of the two horns, as detailed in the HFSS exercise on horn antennas uploaded in October 7.
3. The experimentally measured patterns at 3 frequencies within the respective bands of the two SGHs are obtained after the practical lecture at the DTU-ETC anechoic chambers from November 18.

Taking outset in the above, write a report including the following:

- Brief description of the measurement setups at the DTU-ETC antenna laboratories.
- Analysis of achieved results.
- Comparison of theoretical, simulated and experimental results.

The intention of this exercise is that you develop an understanding of what are the most important points of this comparative analysis. Therefore, the expected content of the report is left intentionally open-ended, and up to your own interpretation. However, some the most important parameters, on which your analysis can be based, are the following: radiation pattern, maximum directivity, side-lobe level, back radiation, cross-polar level, beamwidth, frequency dependence.

The data should be presented in the form of radiation patterns, tables, or other representation formats you deem relevant. The analysis and comparison should not be limited to a description of the data presented in the figures; a good analysis should be able to explain the presented results, supported by the relevant theory where necessary.

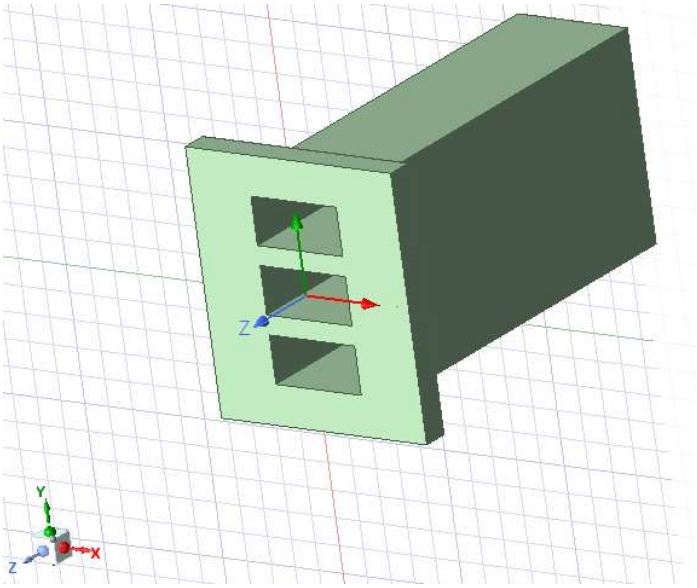
The extension of this exercise should ideally be no more than 15 pages, including figures. Thus, for the sake of brevity, please do not include the following:

- A repeat of the theory of horn antennas given by the Balanis book.
- A repeat of the modelling and simulation process in HFSS.

Part 2: Simulation of an aperture array

The second part of the assignment is a shorter modelling exercise using HFSS. The objective is to simulate a linear *N*-element array of aperture elements, arranged along the y-axis with a separation of $\lambda/2$ at the frequency of 30 GHz. First, we consider a uniformly-fed array, and the directivity of this array should be between 13.5 dB and 15 dB.

- As a starting point, use the open-ended waveguide file *OEWG.aedt* that you have created while working through the HFSS exercise on horn antennas. This open-ended waveguide will be the radiating element of the array.
- To reduce the impact of edge scattering, include a front panel to the array with a thickness of 2 mm, extending beyond the edges of the outer elements. This allows the induced currents at the limits of the array to be close to zero, and reduces the radiation in the backside of the antenna.
- You can use the following figure as a guide for your own design.



Tips:

- *Use the "Duplicate along line" tool to create the array from the initial element.*
- *Keep the size of the radiation box as small as possible to minimize the computation time.*
- *The size of the antenna should in principle be small enough to be solved on the student version of HFSS. If that is not the case, you can run the simulation in the full version HFSS available at the computers in the databar of b348.*

Once you have created your model, answer the following questions as concisely as possible:

1. Write the theoretical array factor of this uniform array. Plot the far-field directivity radiation pattern of the co-polar and cross-polar components on the planes $\varphi = 0$ deg and $\varphi = 90$ deg. What differences do you observe between both patterns?

2. Modify the excitation of the array elements such that the level of the first side lobe is at least 30 dB below the on-axis maximum of radiation. Indicate the values of the excitation coefficients (amplitude and phase). Plot the resulting far-field radiation pattern using markers to indicate the on-axis and first-SLL levels.

Tips:

- *To modify the excitations of the array elements, go to "HFSS -> Fields -> Edit sources…"*
- *Once the design has been solved with HFSS, changing the excitation of the elements does not require a fresh simulation. You can verify that changing the excitation values will instantly modify the radiation patterns of your results.*

3. Modify the excitation of the array elements such that each element is fed with equal amplitude and a progressive phase resulting in a maximum at the direction of $\theta = 30$ deg on the plane of the array. State the values of the phase of each element. Plot the resulting far-field radiation pattern. Compare the values of maximum directivity in this case vs. the case of a broadside array from question 1. What is the difference between the two, and what do you think causes this difference?

Tip: research the concept of scanning loss to answer this last question.